

Demo: Forecasting a Café's Daily Ingredient Demand

A Lightweight Forecasting Pipeline With Damped Triple Exponential Smoothing

Bastian Queckenstedt · Tobias Christopher Kröncke · Efe Enes Nayci

Department of Informatics, Universität Hamburg

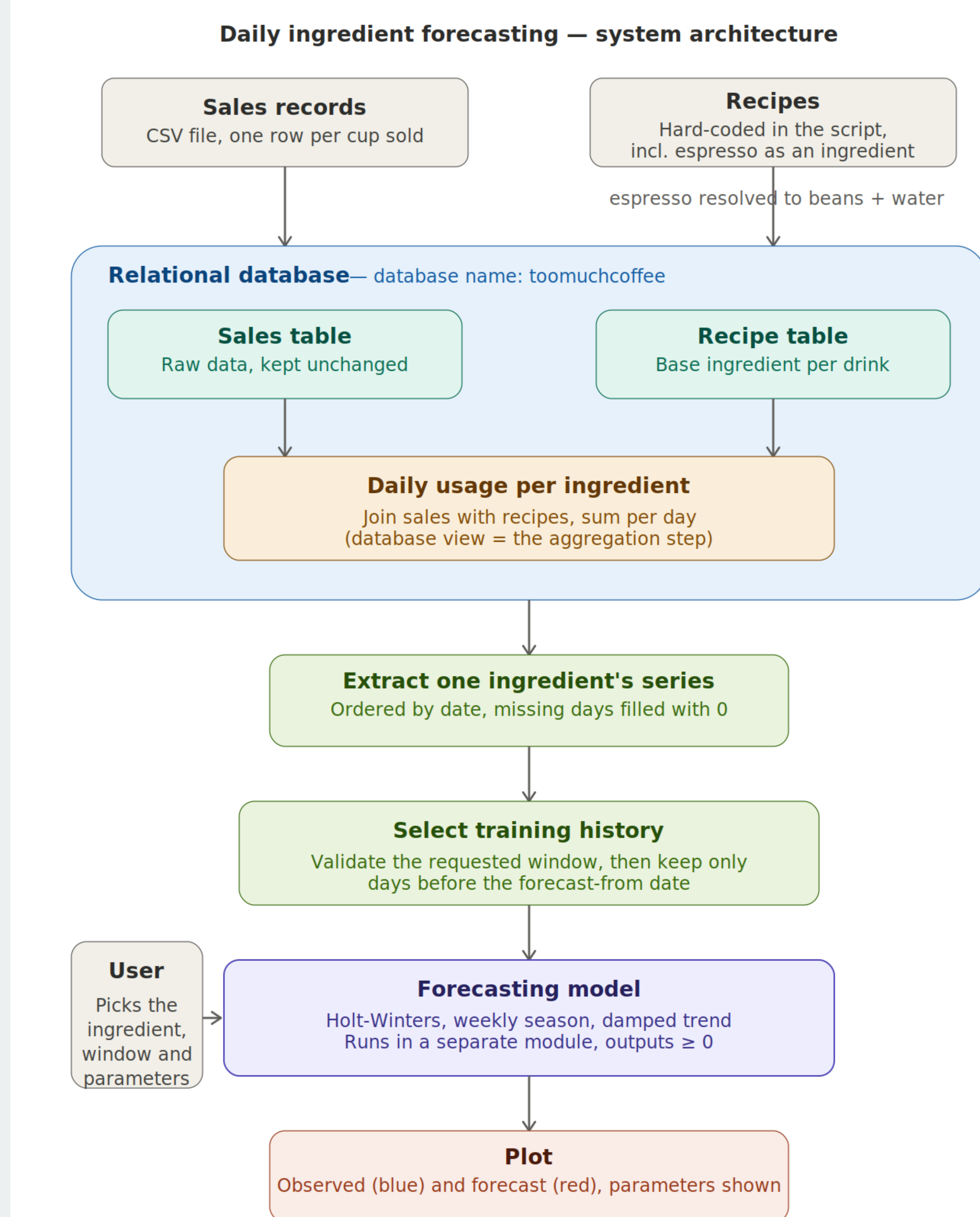
The problem

- A café has records on how many cappuccinos, cocoas, espressos etc. it sold in the past.
- It must purchase the right amounts of what it will consume in the future: **coffee beans, milk, water, cocoa, sugar, salt.**

What the demo does

- One year of real sales (3 547 cups, Mar 2024 – Mar 2025) is loaded **unchanged** into PostgreSQL.
- Recipes are resolved into **six base ingredients** (espresso shots → 18 g beans + 30 ml water; “milk or water” → water).
- A SQL **view** joins sales with recipes and aggregates the daily usage per ingredient – declaratively, inside the database.
- A Python CLI extracts one series, forecasts it, and plots the result automatically.

Architecture



How: damped Holt–Winters

Additive triple exponential smoothing, implemented from scratch, season length $m = 7$ days:

$$\ell_t = \alpha(x_t - s_{t-m}) + (1 - \alpha)(\ell_{t-1} + \phi b_{t-1})$$

$$b_t = \beta(\ell_t - \ell_{t-1}) + (1 - \beta)\phi b_{t-1}$$

$$s_t = \gamma(x_t - \ell_t) + (1 - \gamma)s_{t-m}$$

- Damping $\phi < 1$ lets the trend level off instead of exploding over long horizons.
- Forecasts are clipped at zero – negative usage is impossible.

Try it yourself!

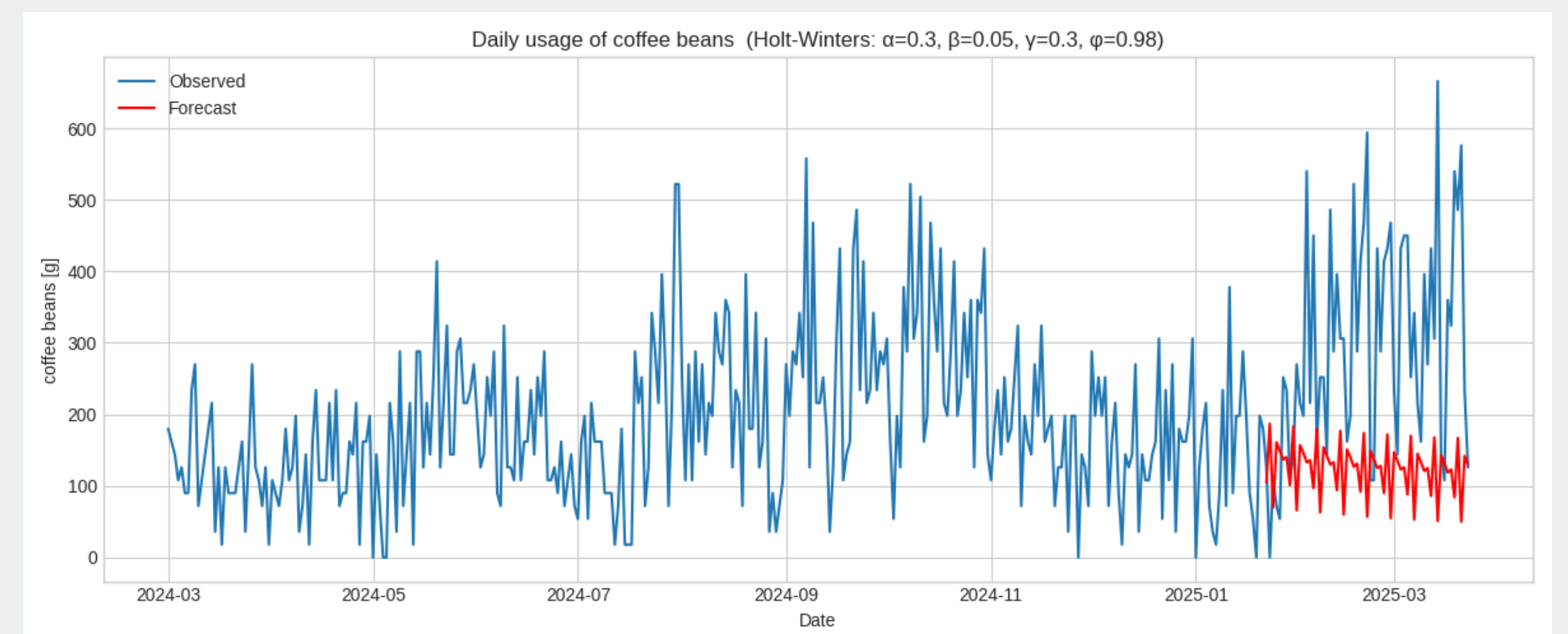
Switch the ingredient, move the window, break the model:

```

$ timeseries.py -i milk
$ timeseries.py -i "coffee beans" -a 0.3 -g 0.3
$ timeseries.py -p 1.0    (undamped trend!)
$ timeseries.py -a 1.0    (chasing noise!)
  
```

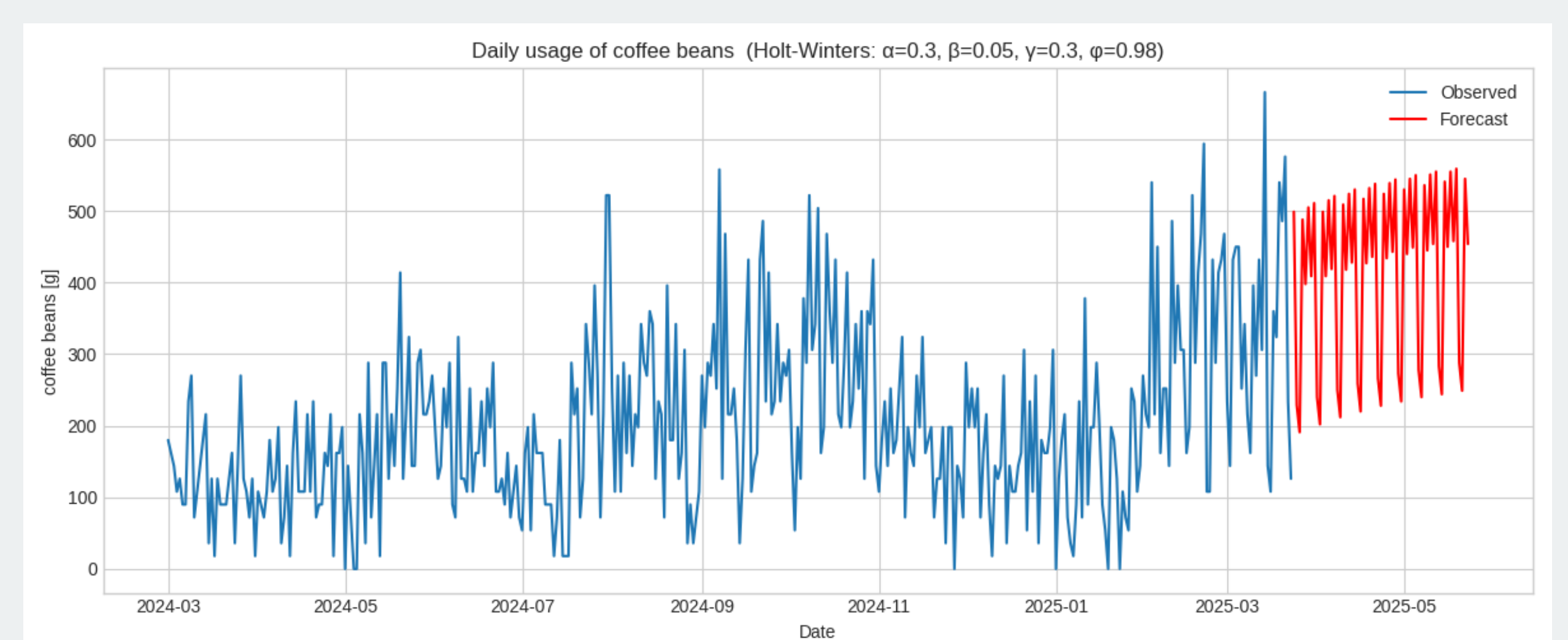
- All four smoothing parameters ($\alpha, \beta, \gamma, \phi$) are command line flags.
- Every run finishes in well under a second.

Does it work? Validation



The model forecasts the **last two observed months** using only the history before the window – the weekly rhythm is clearly reproduced (coffee beans, $\alpha=0.3, \beta=0.05, \gamma=0.3, \phi=0.98$).

Looking ahead



Two-month forecast **beyond the data**: the answer to the café's actual planning question – how many beans to buy next week.